

04-electricity-magnetic-and-heating-effect NCERT Grade 8 Science (Curiosity)

1. Fill in the Blanks

(i) The solution used in a Voltaic cell is called _____.

Answer:-electrolyte

Explanation:* A Voltaic cell consists of electrodes dipped in a liquid solution called an electrolyte.

(ii) A current carrying coil behaves like a _____.

Answer:-magnet (or electromagnet)

Explanation: When an electric current flows through a coil of wire, it generates a magnetic field around it and behaves as a magnet.

2. State True or False

(i) Dry cells are less portable compared to Voltaic cells.

Answer:-False

Explanation: Dry cells contain a moist paste instead of a liquid electrolyte, making them highly leak-proof and much more portable than liquid Voltaic cells.

(ii) A coil becomes an electromagnet only when electric current flows through it.

Answer:-True

Explanation: An electromagnet relies on the magnetic effect of electric current; it loses its magnetism as soon as the current stops flowing.

(iii) An electromagnet, using a single cell, attracts more iron paper clips than the same electromagnet with a battery of 2 cells.

Answer:-False

Explanation: A battery with 2 cells provides a larger current, producing a stronger magnetic field that can attract more clips than a single cell.

3. An electric current flows through a nichrome wire for a short time.

(i) The wire becomes warm.

(ii) A magnetic compass placed below the wire is deflected.

Choose the correct option:

(a) Only (i) is correct

(b) Only (ii) is correct

(c) Both (i) and (ii) are correct

(d) Both (i) and (ii) are not correct

Answer:- (c) Both (i) and (ii) are correct

Reasoning: A current-carrying wire exhibits both the heating effect (making the nichrome wire warm) and the magnetic effect (deflecting the compass needle). its melting point is about 1,400 °C / 2,550 °F)

4. Match the Column

Column A	Column B
(i) Voltaic cell	(a) Best suited for electric heater
(ii) Electric iron	(b) Works on magnetic effect of electric current
(iii) Nichrome wire	(c) Works on heating effect of electric current
(iv) Electromagnet	(d) Generates electricity by chemical reactions

Ans.

Column A	Column B
(i) Voltaic cell	(d) Generates electricity by chemical reactions
(ii) Electric iron	(b) Works on heating effect of electric current
(iii) Nichrome wire	(d) Best suited for electric heater
(iv) Electromagnet	(a) Works on magnetic effect of electric current

5. Nichrome wire is commonly used in electrical heating devices because it:

- (i) It is a good conductor of electricity.
- (ii) It generates more heat for a given current.
- (iii) It is cheaper than copper.
- (iv) It is an insulator of electricity.

Answer:-(ii) generates more heat for a given current

Explanation:* Nichrome offers significantly higher electrical resistance compared to metals like copper. Due to this resistance, it converts a larger fraction of electrical energy into heat energy when current passes through it.

6. Electric heating devices (like an electric heater or a stove) are often considered more convenient than traditional heating methods (like burning firewood or charcoal).

Give reason(s) to support this statement considering societal impact.

Answer: Clean Air & Health: Burning firewood or charcoal produces harmful smoke, toxic gases, and particulate matter that cause respiratory illnesses like coughing or asthma. Electric heaters produce heat without smoke or air pollution.

2. Environmental Protection: Using electricity reduces deforestation (cutting trees for firewood) and carbon emissions associated with burning wood/charcoal.

3. Safety & Speed: Electric appliances can be instantly controlled with switches, temperature dials, and automatic cut-offs, making them cleaner, faster, and safer to operate indoors.

7. Look at Fig. 4.4a.

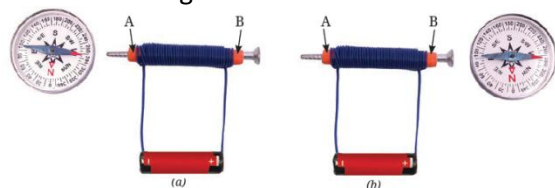


Fig. 4.4: Compass needle near (a) End A; (b) End B of an electromagnet

If the compass placed near the coil deflects:

- (i) Draw an arrow on the diagram to show the path of the electric current.

- (ii) Answer: Electric current flows out from the positive terminal (+) of the cell, pass through the coiled wire around the nail, and returns to the negative terminal (-) of the cell.

(ii) Explain why the compass needle moves when current flows.

Answer: Electric current flowing through the coil produces a magnetic field around it, turning the coil into an electromagnet. The magnetic compass needle (which is a small magnet) detects this field and gets deflected.

(iii) Predict what would happen to the deflection if you reverse the battery terminals.

Answer: Reversing the battery terminals reverses the direction of the current. Consequently, the magnetic poles of the electromagnet swap places (North becomes South and vice-versa), causing the compass needle to deflect in the opposite direction.

8. Suppose Sumana forgets to move the switch of her lifting electromagnet model to OFF position. After some time, the iron nail no longer picks up the iron paper clips, but the wire wrapped around the iron nail is still warm. Why did the lifting electromagnet stop lifting the clips? Give possible reasons.

Answer: The possible reasons are:

1. Cell Exhaustion ("Dead Cell"): Leaving the circuit permanently 'ON' drains the chemical energy of the electric cell rapidly. Once the cell is drained, no electric current flows, the magnetic field disappears, and the nail stops attracting paper clips.
2. Increased Wire Resistance Due to Heating: Continuous current flow causes heating of the wire due to resistance. As the wire heats up, its resistance increases, further reducing the magnitude of the electric current and weakening the magnetic strength.

9. In Fig. 4.12, in which case the LED will glow when the switch is closed?

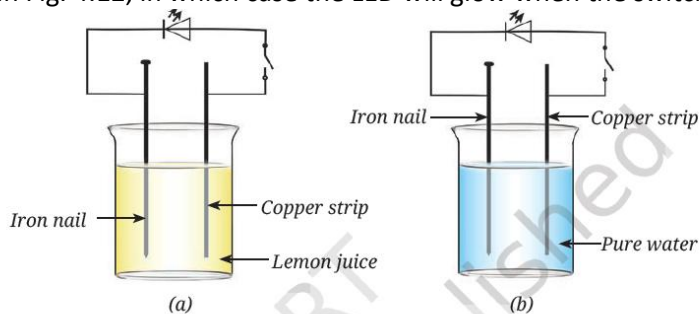


Fig. 4.12

Case (a): Iron nail + Copper strip in Lemon juice

Case (b): Iron nail + Copper strip in Pure water

Answer: Case (a)

Explanation: Lemon juice contains citric acid which acts as an electrolyte to conduct electricity and create a chemical reaction between the iron nail and copper strip (a Voltaic cell). Pure water lacks dissolved salts/acids, so it does not act as an effective electrolyte, and current will not flow to light up the LED.

10. Neha keeps the coil exactly the same as in Activity 4.4 but slides the iron nail out, leaving only the coiled wire. Will the coil still deflect the compass?

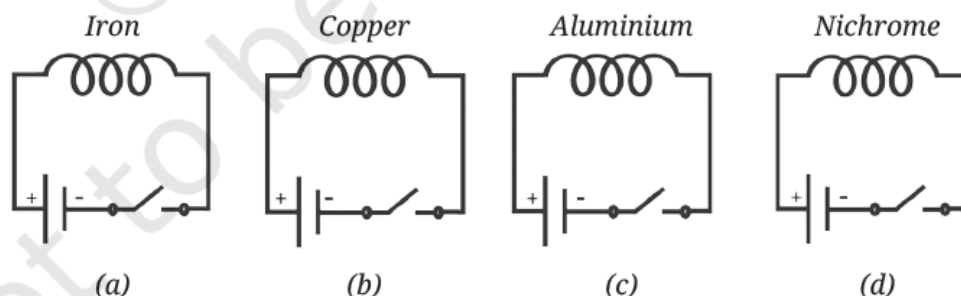
If yes, will the deflection be more or less than before?

Answer:

Yes, the coil will still deflect the compass. The deflection will be LESS than before.

Reasoning: A current-carrying coil inherently creates a magnetic field. However, inserting a soft iron core inside the coil greatly concentrates and strengthens the magnetic field. Removing the iron nail weakens the magnetic field, resulting in a smaller deflection of the compass needle.

11. We have four coils of similar shape and size, made up from iron, copper, aluminum, and nichrome



(Fig. 4.13).

Fig. 4.13

When current is passed through the coils, compass needles placed near the coils will show deflection:

- (i) Only in circuit (a)
- (ii) Only in circuits (a) and (b)
- (iii) Only in circuits (a), (b), and (c)
- (iv) In all four circuits

Answer:(iv) In all four circuits**

Explanation: Hans Christian Oersted established that an electric current flowing through any electrical conductor produces a magnetic field around it, regardless of the conductor material (iron, copper, aluminium, or nichrome). Therefore, current passing through any of these coils will cause a nearby magnetic compass needle to deflect.